

Experiment 3**Static Structural Analysis of Fixed Beam**

Reg. No. 2023111073	Date 08-09-2026
Software ANSYS Workbench (Student)	Beam 500 × 20 × 10 mm

Aim

To perform static structural analysis of a fixed beam with the application of a distributed load and a point load, applied separately at its top surface.

Software Required

ANSYS Workbench (Student / Academic Version)

Problem Statement

Analyse the deformation and stress distribution of a fixed beam subjected to an external load using ANSYS Workbench.

Theory

Static structural analysis is used to determine the stresses, strains, and deformation caused by applied loads.

Procedure

- 1 Open ANSYS Workbench.
- 2 Drag and drop a Static Structural system into the Project Schematic.
- 3 Double-click Geometry and create a rectangular beam using SpaceClaim / DesignModeler.
- 4 Set beam dimensions: Length = 500 mm, Width = 20 mm, Height = 10 mm.
- 5 Close and save the geometry.
- 6 Open Model and assign Structural Steel as the material.
- 7 Generate the mesh using default settings.
- 8 Apply a fixed boundary condition at both ends of the beam.

- 9 Apply a distributed load of 5000 N/m-length downward at the top surface.
- 10 Solve the model.
- 11 Insert Total Deformation and Equivalent (von-Mises) Stress, save the result pictures, and note down the results.
- 12 Change to a 2000 N point load at the centre, re-solve, and measure the maximum deformation and von-Mises stress. Record the new results.

Results

The maximum deformation and maximum bending stress occur at the centre of the beam for both Case A (distributed load) and Case B (point load at the centre).

Total Deformation

The value of maximum deformation for each case is given below, with the corresponding contour plots.

Case A — Distributed Load (5000 N/m)

Maximum deformation: 0.0024345 m (2.4345 mm)

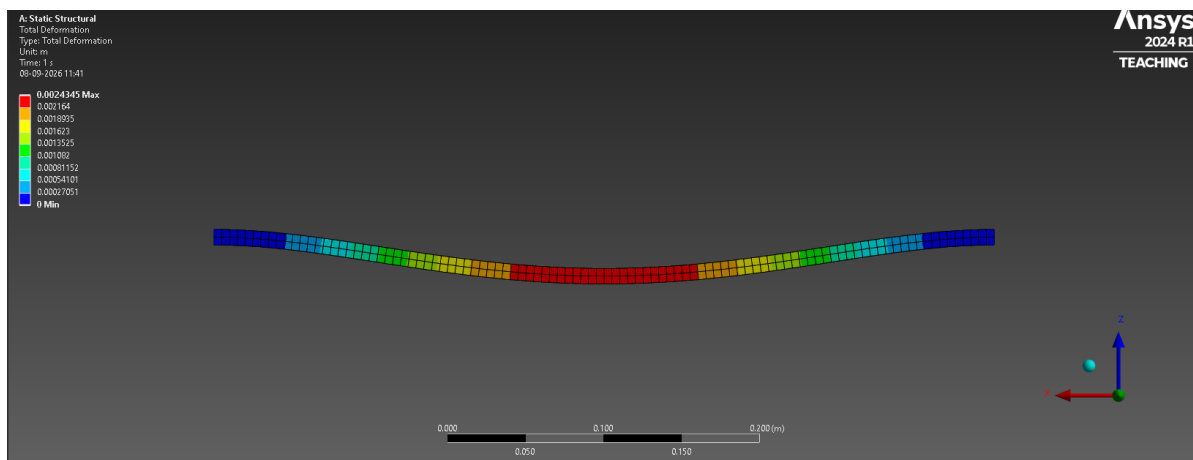


Fig. 1 — Total Deformation, Case A (Distributed Load)

Case B — Point Load (2000 N at centre)

Maximum deformation: 0.0038952 m (3.8952 mm)

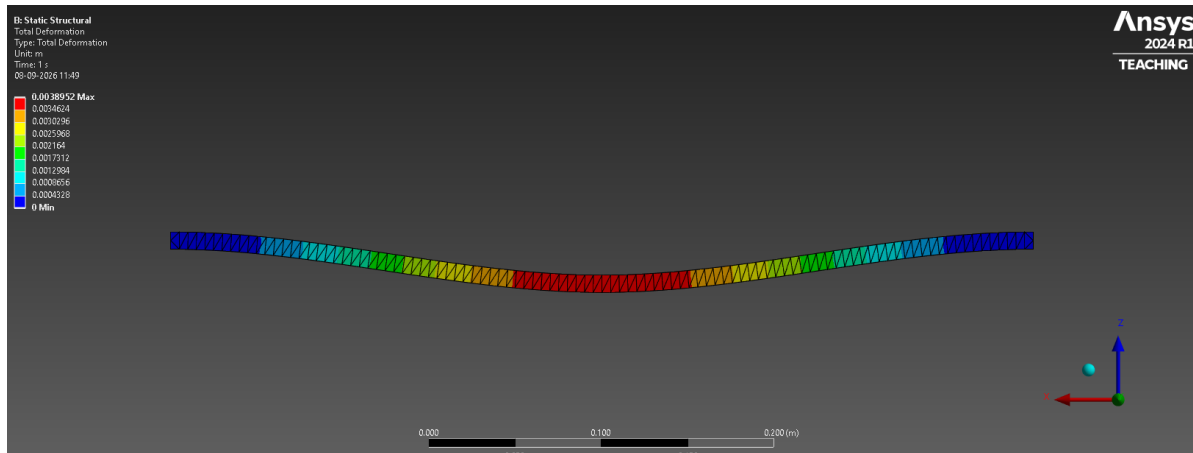


Fig. 2 — Total Deformation, Case B (Point Load)

von-Mises Stress

The value of maximum stress for each case is given below, with the corresponding contour plots.

Case A — Distributed Load (5000 N/m)

Maximum stress: 3.0536×10^8 Pa (305.36 MPa)

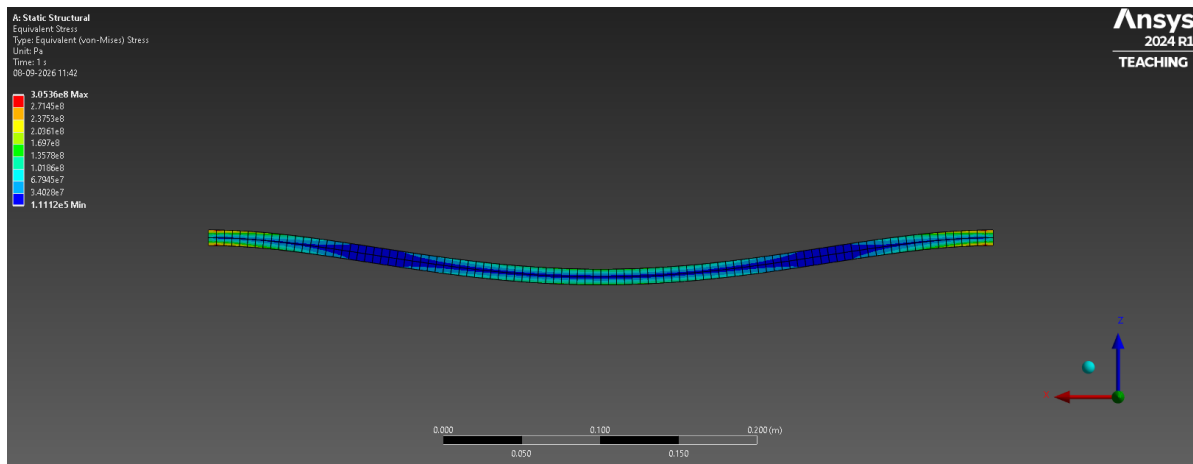


Fig. 3 — Equivalent (von-Mises) Stress, Case A (Distributed Load)

Case B — Point Load (2000 N at centre)

Maximum stress: 3.6669×10^8 Pa (366.69 MPa)

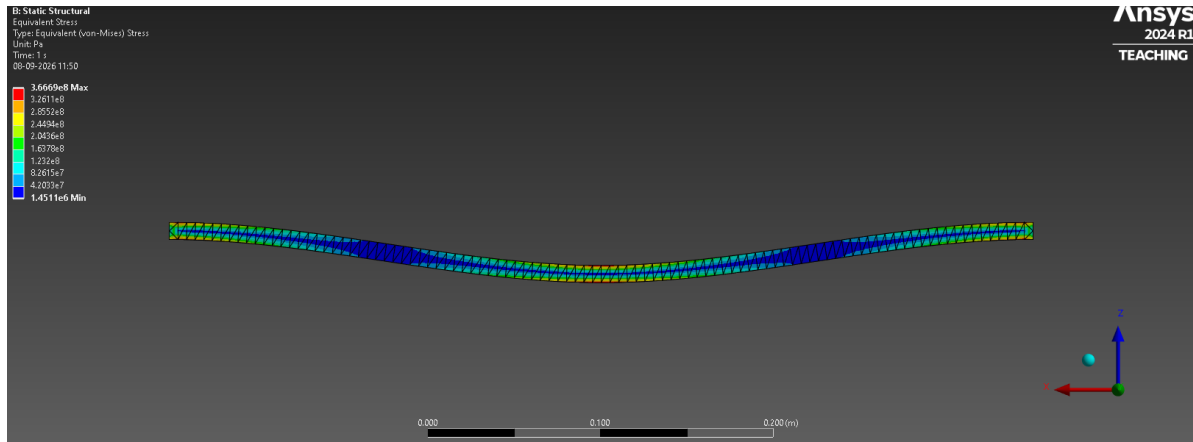


Fig. 4 — Equivalent (von-Mises) Stress, Case B (Point Load)

Summary of Results

Load Case	Max. Total Deformation	Max. von-Mises Stress
Case A — Distributed load (5000 N/m)	2.4345 mm	305.36 MPa
Case B — Point load (2000 N, centre)	3.8952 mm	366.69 MPa

Conclusion

Static structural analysis of a fixed beam was successfully performed, and the deformation and stress distributions were visualised for both the distributed-load and point-load cases.

The point load at the centre produced a higher maximum deformation and a higher peak von-Mises stress than the equivalent-intensity distributed load, consistent with the more concentrated nature of the loading.